

Zapping to death!

DARPA's new laser research boosts Airborne Death Rays. Read more at: <http://dgit.in/1Aq7xgK>



Big but light

Saturn, the ringed planet, is light enough to float on water (If you can find a tub that's big enough!)

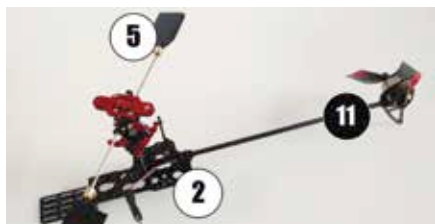
DO



The outer body of the craft



The inner setup of the main motor



The tail assembly



The rotor head and electronic speed controllers

well as the mechanical components. **(3) Skid landing gear:** Crafted out of compliant shock absorbing material, this reduces the risk of expensive crashes during landing manoeuvres. **(4) Main rotor:** A pair of blades angled sharply to generate lift. They are mounted on a mechanically complicated assembly that allows a fine control over their angle of attack (pitch) to generate varying amount of lift. **(5) Flybar:** The second set of blades found on fixed-pitch helicopters. These are mechanically linked to the main rotor and automatically maintain and correct its pitch. **(6) Rotor head:** An extremely complicated mechanical solution that translates flight-control input

into precise changes to the main rotor orientation and pitch. It's instrumental in controlling helicopters. **(7) Main motor:** Drives the main rotor to generate lift. **(8) Electronic Speed Controllers (ESC):** Electronics essential for achieving commutation and speed control in brushless motors. **(9) Receiver/Control Unit:** Incorporates the radio that receives user input signal from the transmitter. In micro RC helicopters, this also incorporates microcontrollers responsible for Collective Cyclic Pitch Mixing (CCPM). CCPM allows user inputs to be translated into complicated servo commands needed to achieve desired flight control. The assembly also houses gyroscope(s), which is crucial to maintain stability. **(10) Servomechanism:** Electronic control system that's connected to the rotor head by pushrods. These are responsible for conveying user inputs into adjustments to the rotor head to achieve proper flight control. **(11) Tail assembly:** Consists of the tail boom, motor, rotor, and rudder. This essentially provides compensating torque to nullify the natural torque generated by the main rotor.

Now that you know what makes an RC helicopter tick, it's time to decide which type of single-rotor RC helicopter you prefer. Choosing the smaller micro RC form factor brings down cost considerably because the gyroscope is integrated into the main receiver/control unit. The reduced weight and a relatively small amount of torque produced by the main rotor also mean that you can get away with much simpler and hence cheaper tail rotor assembly. Modernisation of the hobby has made some of the choices redundant. The widespread adoption of the superior, interference-free 2.4GHz radios rendering glitchy FM radios obsolete is a prime example. However, fundamental options such as the binary choice between fixed and collective pitch helicopters still remains the most important consideration.

Fixed or Collective Pitch?

A fixed pitch implementation employs a flybar assembly to automatically control the pitch angle of the main rotor. It inherently makes the helicopter self-correcting

and hence easy to fly. But it also reduces the responsiveness and overall manoeuvrability as a consequence. On the other hand, collective pitch helicopters are capable of what's known as 3D flight – the ability to perform gravity-defying manoeuvres such as inverted flight, barrel rolls, and multiple loops. However, the lack of flybar-assisted dampening makes them twitchy and harder to control for beginners. The additional degree of motion also calls for sophisticated six-axis gyroscopes that increase cost.

A fixed pitch helicopter requires a four-channel radio controller, where each radio channel corresponds to conveying input signals to a motor, actuating servomechanism, or manipulating any electronic switch. In a four-channel setup, these channels control (1) main motor for generating lift through the main rotor, (2) tail rotor to control yaw, (3) fore and aft cyclic pitch through the rotor head, and (4) left and right cyclic roll again using the rotor head. Collective pitch versions need more expensive six-channel radios for additional pitch and gyro gain/setting control, apart from an extra servo mechanism for controlling the main rotor pitch angle.

Brushless or Coreless Motors?

Deciding between regular permanent magnet DC motors, coreless, or brushless variants is the next most important choice. Regular DC motors are good enough for RC cars, but they can neither handle the high amperage values of RC helicopter circuits nor do they possess a sufficient power-to-weight ratio. As the name suggests, the rotor of a coreless motor lacks an iron core, making it incredibly lightweight. The reduced inertia allows it to accelerate faster and achieve a significantly higher RPM than a regular DC motor. Despite their advantages, coreless motors employ brushes to transfer current to a rapidly spinning armature. This reduces efficiency and limits both power and torque capability.

Brushless motors, on the other hand, involve permanent magnets moving around a fixed armature. This is a highly efficient arrangement that doesn't depend